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Civil Engineering

AIRCRAFT FIRE PROTECTION FOR EXERCISES AND CONTINGENCY RESPONSE OPERATIONS

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This pamphlet provides planners guidance for contingency response operations lasting a maximum of 120 days. Planners use this guidance to determine the minimum number of Fire Emergency Services (FES) aircraft rescue and fire fighting (ARFF) manpower and vehicles necessary to provide fire suppression for Air Force (AF) aircraft during major exercises and real world contingencies. These operations may include exercises for training purposes, disaster response, humanitarian relief operations, or other non-combat type operations. Determining requirements is based on the type and size of aircraft being protected. For the purposes of this document, applicable AF ARFF vehicle sets and requirements are found in Allowance Standard (AS) 010, *Vehicle Fleet (Registered) All MAJCOM Common*, and are based on National Fire Protection Association (NFPA) Standard 403, *Aircraft Rescue and Fire Fighting Services at Airports,* and AF NFPA 403, *Technical Implementation Guide.* These sets are used as a baseline and modified to meet exercise and contingency requirements. The capabilities outlined in this pamphlet provide ARFF fire suppression and limited rescue capability, but do not include structural fire protection or other emergency response capabilities. This guidance applies to all AF Active, Reserve, and Air National Guard (ANG) Civil Engineer (CE) units. Ensure that all records created as a result of the processes prescribed in this publication are maintained in accordance with (IAW) Air Force Manual (AFMAN) 33-363, *Management of Records*, and disposed of IAW the AF Records Disposition Schedule located at **https://www.my.af.mil/gcss-af61a/afrims/afrims**. Use AF Form 847, *Recommendation for Change of Publication*, to submit any recommended changes, clarification requests, or command supplements to this pamphlet to the Air Force Civil Engineer Center (AFCEC/CXF), 139 Barnes Drive, Suite 1, Tyndall Air

Force Base, FL. 32403-5319. Forms may be electronically forwarded to AFCEC/CXF Corporate Mailbox, **AFCEC.CXF.WORKFLOW@us.af.mil.**

SUMMARY OF CHANGES

This document has been substantially revised and must be completely reviewed. This revision addresses changes contained in Air Force Instruction (AFI) 32-2001, *Fire Emergency Services (FES) Program* and provides consistency with WMP 1.

1. Scope. This pamphlet provides guidance for operations lasting a maximum of 120 days and is used for small scale operations. It does not apply to long term ongoing operations or wars. FES guidance for those requirements is found in AFI 32-2001 and WMP 1, *CE Supplement, Appendix 5.*

2. Responsibility. The aircraft user provides ARFF capability at locations other than established AF installations. If the aircraft user cannot provide the minimum ARFF capability listed in Table 1, the user should request support through Air and Space Expeditionary Force Operations or the aircraft provider's Major Command (MAJCOM) Director of Operations (DO). In this case the user may also be responsible for providing funding.

3. Risk Management (RM). Fire fighting capability is dependent on three primary resources: fire fighting vehicles, agent, and personnel. The assumption is that only one major fire incident will occur at a time. This document approaches RM from a perspective of requiring the local risk managers to determine the acceptable levels of risk based on local risk factors. Generally, local risk factors include historical fire experience (if available) and the type and duration of the operation. Refer to Air Force Pamphlet (AFPAM) 90-803, *Risk Management Guidelines and Tools,* for guidance on conducting a risk assessment.

3.1. The maximum length of time this document covers is 120 days in any one-year period. For operations that exceed 120 days, arrangements must be made for permanent FES capability as prescribed in AFI 32-2001 (permanent installations) or WMP 1 (deployed locations).

3.2. In principle, the shorter the exposure time, the lower the risk. Theoretically, operations of very short duration present lower risk than operations approaching 120 days. This exposure time should be a factor when determining the FES resources required.

3.3. ARFF capability is dependent on vehicles, agent and personnel available to conduct operations at an incident scene. Capability for each primary resource (vehicles, agent and personnel) is described as Optimum Level of Service (OLS), Reduced Level of Service (RLS), Critical Level of Service (CLS) or Inadequate Level of Service (ILS). Overall FES capability is the lowest level of service available for either resource. For example, when agent is OLS and personnel are CLS, the overall capability is CLS.

3.4. The fundamental purpose of fire fighting operations is to save lives and property. Therefore, the resources necessary to deliver the various levels of service are calculated based on scenarios that may be encountered. OLS is the highest level of service providing the capability to perform rescue from a large fire (exterior and interior) involving an aircraft. Resources are provided to knockdown, extinguish or control the exterior fire and fight interior fires enabling rescue to be performed. OLS provides the highest probability of

success. Conversely, if CLS is provided fire crews would not be expected to perform rescue from a large fire scenario. They may be successful in extinguishing a small fire on one side/one location or perform rescue from a fighter-type aircraft. However, these operations would have to be prioritized since they would not have capability to perform these operations simultaneously.

3.5. Saving lives is the primary objective. The probability of saving property diminishes as the fire grows. Fire crews focus on early intervention so they can intervene in time to avoid major damage. When early intervention does not occur it is not always possible to save property.

4. Planning Guidance. Local risk managers determine the quantity of agent required based on the risk assessment discussed in paragraph 3 above. The quantities discussed in this document may be less than that available at a permanent installation due to the short exposure time (less than 120 days).

4.1. The amount of agent required is based on the type and size of aircraft. Aircraft are grouped into 6—ARFF Vehicle Sets described in Table 1. Fire fighting agent is delivered by ARFF vehicles with capacities of 500, 1,000, 1,500, 3,000 or 3,300 gallons. Table 1 indicates the quantity of agent required for various levels of service. Ideally the quantity of agent required to provide OLS should be provided; however, for short-duration operations RLS or CLS may be provided with the appropriate notices to Airmen and waivers.

4.1.1. The Fire Chief or Senior Fire Officer (SFO) should discuss the risk associated with each level of service with the Commander and ensure the risk they are accepting is fully understood.

4.1.2. OLS is when all authorized resources are available for emergency response within response time standards. OLS provides sufficient capability for quick response and sustained operations after arrival on scene. During OLS, emergency response forces shall accomplish all objectives when responding to emergency incidents.

4.1.3. RLS is when the Emergency Response Capability (ERC) is less than OLS but greater than CLS. Sufficient capability is provided for initial response, scene assessment and implementation of mitigation tactics. This level of service represents increased risk/loss potential due to lack of ERC to perform rescue and sufficient mitigation tactics simultaneously. FES objectives may not be successful during situations where simultaneous rescue and fire fighting activities are required.

4.1.4. CLS capability exists when 7 firefighters are available to respond to an emergency within the response time standards. Upon arrival, the Incident Commander will determine the appropriate actions to be taken depending upon their initial evaluation of the situation. Successful outcomes can only be expected when the incident can be quickly mitigated. Firefighters are expected to revert to defensive operations when the emergency cannot be quickly contained. This level of service represents limited rescue capability and increased risk/loss potential due to limited resources. Due to exhaustion and resource limitations, CLS is considered a “one shot” capability that cannot be sustained for more than approximately 15 minutes.

4.1.5. ILS capability represents an amount of agent less than CLS and the property involved in fire is expected to be destroyed. Except at fires in the incipient stage, entry for fire fighting purposes is not permitted.

4.2. Regardless of available manpower, CLS agent in Table 1 represents increased risk to aircrews and reduces the probability that the fire can be extinguished. The RLS column is a safer alternative and provides increased capability. CLS is the minimum level of agent acceptable and should be limited to very short duration operations. This level of protection is necessary before operations begin.

4.3. See Attachment 2 for actual Department of Defense (DoD) Aircraft Quantity (Q)1, Q2, and Q3 Factor Calculations. Actual Q Factor Calculations should also be considered when developing RM plans or making risk-based decisions. For more information regarding RM see AFI 32-2001.

Table 1. Fire Fighting Personnel/Agent Requirements.

Aircraft Type		Optimum Level of Service		Reduced Level of Service		Critical Level of Service		Inadequate Level of Service	
Aircraft	AF Vehicle Set	OLS - Firefighters	OLS - Gallons Q1+Q2+Q3	RLS - Firefighters	RLS -Gallons Q1+Q2	CLS - Firefighters	CLS - Gallons Q1	ILS - Firefighters	ILS - Gallons
A-10, C-21, F-15, F-16, F-22, F-35, F-117, T-37B, BQM-34, MQ-1A/B, T-38, AT-38, MQM-107, T-6A, UV-18, QF-4, CV-22, UH-1N, C-38A, T-1, RQ-4, and C-12	1	14	2,500 - 1,340	13 - 8	1,339 - 513	7	512 - 325	6 or below	324
C-20	2	14	4,000 - 2,760	13 - 8	2,759 - 1,316	7	1,315 - 752	6 or below	751
C-9, C-22, C-32, C-37, C-40, C-130, E-3, E-8, T-43, MH-53 and RC-135	3	14	5,000 - 4,880	13 - 8	4,879 - 3,027	7	3,026 - 1,322	6 or below	1,321
B-1, B-2, B-52, C-17, KC-46 and KC-135	4	16	8,000 - 7,780	15 - 8	7,779 - 4,364	7	4,363 - 1,732	6 or below	1,731
E-4 (747), KC-10 and VC-25	5	17	10,000 - 9,570	16 - 8	9,569 - 6,292	7	6,291 - 2,330	6 or below	2,329
C-5	6	18	13,000 - 12,626	17 - 8	12,625 - 7,508	7	7,507 - 2,589	6 or below	2,588
* Firefighter numbers are on a per shift basis									
** Below Optimum Level of Service – Aircrew Awareness (NOTAM)									
** Below Reduced Level of Service – Mission Commander or OG/CC approval									
** At or Below Critical Level of Service – Waiver approval as specified in paragraph 7.1									

4.4. FES planners should select the largest vehicles available to minimize the number of drivers required to operate them.

4.5. Recommend 6 firefighters per operational period with two P-19s or Rapid Intervention Vehicle (RIV)s for contingency airfields supporting Category 1 Remotely Piloted Aircraft (RPA) operations. The absence of a rescue requirement warrants the reduced capability.

4.6. Installation or deployed Commander, Air Force Forces should consult with the MAJCOM or Area of Responsibility (AOR) Fire Chief to establish capabilities for operations not addressed in this document.

4.7. A minimum of two ARFF vehicles are recommended for all locations in case of vehicle failure.

4.8. Required agent may be provided in water tankers so long as at least one ARFF vehicle with turret capability is also provided.

4.9. When operating at locations with multiple types of aircraft the largest aircraft determines the ARFF requirement.

4.10. When the required fire fighting capability is temporarily not available due to unavoidable circumstances for periods more than 72 hours, the Fire Chief will develop a RM plan and provide it to the responsible Commander for their review and approval.

5. Personnel Requirements. The previous paragraphs discussed the fire fighting agent required. This section addresses the manpower necessary to accomplish tasks in order to achieve OLS, RLS and CLS.

5.1. Note: The personnel numbers listed in Table 1 would constitute firefighters working 24 hours a day/7 days a week. If flying operations increase, personnel numbers listed in Table 1 may need to be increased. The Fire Chief or SFO will determine how firefighters will be utilized.

5.2. If structural fire protection is required, consult with the AOR or MAJCOM Fire Chief to provide appropriate structural fire protection requirements. This can be provided by adding structural fire fighting vehicle(s). Cross-manning (staffing) should be used to minimize the need for additional personnel.

6. Exceptions. The responsible commander may exclude fire protection for infrequent flying operations including:

6.1. USAF Vehicle Sets 1 through 3: Not more than four takeoffs and four landings within seven consecutive days.

6.2. USAF Vehicle Sets 4 through 6: Not more than two takeoffs and two landings within seven consecutive days.

6.3. AFI 11-299, *Nuclear Airlift Operations*, requires a minimum of one AF P-19 or equivalent and manpower for locations where aircraft carrying special weapons are operated, loaded or unloaded. This is required regardless of the number of takeoffs and landings occurring.

7. Waivers. When protection capability is at or below CLS, the airlift provider MAJCOM/DO or ANG/DO must grant waivers for operations. For exercises and contingency operations, the Director of Mobility Forces and the Joint Force Air Component Commander must waive ARFF requirements. When time does not permit formal (written) waiver action, the MAJCOM/DO (waiver authority) files an after-action summary. The MAJCOM/DO provides information copies of all waivers to the MAJCOM Fire Protection Office. A waiver shall include:

7.1. Date(s) and type of operations.

7.2. Type aircraft involved.

7.3. Description of available ARFF assets to include dedicated manpower, amount of fire fighting agent available, and number of ARFF vehicles.

7.4. Mission Impact Statement. A mission impact statement must accompany all waiver requests.

NOTE: No waiver is required for the first aircraft in and the last aircraft out carrying ARFF equipment.

JUDITH A. FEDDER, Lt Gen, USAF
DCS/Logistics, Installations, and Mission Support

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 11-299, *Nuclear Airlift Operations*, 19 March 2008

AFI 32-2001, *Fire Emergency Services Program,* 27 February 2014

AFMAN 33-363, *Management of Records*, 1 March 2008

AFPAM 90-803, *Risk Management Guidelines and Tools*, 11 February 2013

AS 010, *Vehicle Fleet (Registered) All MAJCOM Common,* 2008

WMP-1*, CE Supplement, Appendix 5*, Jan 2014

NFPA 403, *Standard for Aircraft Rescue and Fire Fighting Services at Airports,* Edition 2014

AF NFPA 403, *Technical Implementation Guide,* 18 October 2010

Adopted Forms

AF Form 847, Recommendation for Change of Publication

Abbreviations and Acronyms

AEFO—Air and Space Expeditionary Force Operations

AFCEC—Air Force Civil Engineer Center

AFI—Air Force Instruction

AFPAM—Air Force Pamphlet

AFRIMS—Air Force Records Information Management System

AOR—Area of Responsibility

ARFF—Aircraft Rescue and Fire Fighting

AS—Allowance Standard

CE—Civil Engineer

CLS—Critical Level of Service

FES—Fire Emergency Services

IAW—In Accordance With

ILS—Inadequate Level of Service

NFPA—National Fire Protection Association

OLS—Optimum Level of Service

PCA—Practical Critical Area

RDS—Records Disposition Schedule

RIV—Rapid Intervention Vehicle

RLS—Reduced Level of Service

RM—Risk Management

RPA—Remotely Piloted Aircraft

SFO—Senior Fire Official

TCA—Theoretical Critical Area

UHP—Ultra High Pressure

WMP—War Mobilization Plan

Terms

Aircraft Rescue Fire Fighting (ARFF) Vehicle— A vehicle intended to transport personnel and equipment to the scene of an aircraft emergency for the purpose of rescuing occupants and conducting rescue and fire fighting operations. ARFF vehicles are equipped with at least one external turret that enables rapid agent application without exposing personnel.

Critical Level of Service (CLS)— The absolute minimum level of FES capability available at a given time. This level of service is only acceptable for short durations and should not be diminished.

Inadequate Level of Service (ILS)— The level of service below CLS. This level of service is unacceptable for fire fighting operations.

National Fire Protection Association (NFPA)— An international organization that publishes codes, standards, recommended practices, and guides through a consensus process approved by the American National Standards Institute.

Optimum Level of Service (OLS)— The maximum level of service that can be continuously provided. During OLS, fire fighting forces are capable of providing all services continuously throughout an event with reasonable expectation of successful offensive fire attack operations, search and rescue, and property conservation.

P18 —– 2,000-gallon water tanker. Normally used for resupply of ARFF vehicles at locations without adequate water supplies. Also provides limited structural fire fighting capability.

P19 —– 1,000/1,500-gallon agent capacity ARFF vehicle.

P22 —– Two wheel drive structural fire fighting pumper. Normally used to provide structural fire fighting capability at locations with adequate water supplies or hydrant systems.

P23 —– 3,000/3,300-gallon agent capacity ARFF vehicle.

P24 —– Four wheel drive structural fire fighting pumper. Normally used to provide structural fire fighting and Wildland Urban Interface fire fighting capability at locations with adequate water supplies or hydrant systems.

P26 —– 4000/5000-gallon water tanker. Normally used for resupply of ARFF vehicles at locations without adequate water supplies. Also provides limited structural fire fighting capability.

Practical Critical Area— Represents the size of large fires that most frequently occurs, usually two-thirds of the TCA for any given aircraft..

P-34 RIV— 500 gallon vehicle with Ultra High Pressure (UHP) fire fighting technology.

Quantity 1 (Q1)— Water requirements to control an exterior fire for one minute.

Quantity 2 (Q2)— Water required to maintain control or extinguish the remaining exterior fire or both.

Quantity 3 (Q3)— Water required for interior fire fighting operations.

Rapid Intervention Vehicle— 500 gallon equivalent ARFF vehicle with UHP fire fighting technology.

Reduced Level of Service (RLS)— The level of FES capability that exceeds the critical but is less than the optimum level of service.

Risk Management— The technique or profession of assessing, minimizing, and preventing accidental loss to a business or mission, as through the use of insurance, safety measures, etc.

Theoretical Critical Area— An area based on the length and width of the aircraft fuselage.

Attachment 2

DOD AIRCRAFT Q FACTOR CALCULATIONS

US Air Force Q Factor Calculations by Type Aircraft and Category.

NFPA Category by Aircraft Type 1-4 Airports	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
T-37B (Tandem)	29 FT 3 IN	5 FT 0 IN	1258	830	1
BQM-34	28 FT 5 IN	1 FT 2 IN	1141	753	1
RQ-1A/B	27 FT 0 IN	3 FT 7 IN	1150	759	1
T-38A	25 FT 3 IN	5 FT 0 IN	1111	733	1
AT-38B/C	25 FT 3 IN	5 FT 0 IN	1111	734	1
MQM-107	18 FT 0 IN	1 FT 0 IN	720	475	1
T-6A	33 FT 4 IN	5 FT 0 IN	1457	968	2
UV-18	51 FT 5 IN	5 FT 9 IN	2660	1756	3
QF-4	58 FT 3 IN	5 FT 0 IN	2971	1961	3
CV-22	57 FT 4 IN	9 FT 0 IN	3153	2081	3
UH-1N	57 FT 3 IN	8 FT 0 IN	3092	2040	3
C-38A	55 FT 7 IN	7 FT 2 IN	2955	1950	3
A-10 & OA-10	53 FT 4 IN	5 FT 0 IN	2720	1795	3
F-16C/D	49 FT 5 IN	5 FT 0 IN	2520	1663	3
T-1A	48 FT 8 IN	5 FT 0 IN	2482	1638	3
C-21A	48 FT 7 IN	4 FT 11 IN	2473	1631	3
RQ-4A	48 FT 5 IN	4 FT 8 IN	2452	1619	3
C-12	43 FT 9 IN	4 FT 6 IN	2209	1458	3
F-35A/B/C	50 FT 8 IN	5 FT 0 IN	2584	1705	3
F-15A/B/C/D	63 FT 9 IN	5 FT 0 IN	3389	2567	4
F-15E	63 FT 9 IN	5 FT 0 IN	3389	2567	4
F-117A	65 FT 11 IN	5 FT 0 IN	4021	2654	4
F-22A	62 FT 1 IN	5 FT 0 IN	3787	2499	4
NFPA Category 5 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
C-20A/B/C/H	83 FT 2 IN	7 FT 4 IN	8759.24	5781.1	5

NFPA Category 6 & 7 Airports by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
C-9A/C (DC-9)	119 FT 3 IN	10 FT 1 IN	12889	8506	6
C-40C (737)	110 FT 4 IN	11 FT 6 IN	11943	7883	6
EC-130E	100 FT 6 IN	10 FT 3 IN	10879	7180	6
T-43A (737)	100 FT 0 IN	11 FT 6.5 IN	10954	7230	6
WC-130H	99 FT 4 IN	10 FT 3 IN	10752	7097	6
HC-130P/N	98 FT 9 IN	10 FT 3 IN	10690	7055	6
MC-130P	98 FT 9 IN	10 FT 3 IN	10690	7055	6
AC-130H/U	97 FT 9 IN	10 FT 3 IN	10581	6984	6
C-130 E/H/J/J-30	97 FT 9 IN	10 FT 3 IN	10581	6984	6
EC-130H	97 FT 9 IN	10 FT 3 IN	10581	6984	6
LC-130	97 FT 9 IN	10 FT 3 IN	10581	6984	6
MC-130E/H	97 FT 9 IN	10 FT 3 IN	10581	6984	6
C-37A	96 FT 5 IN	7 FT 4 IN	10155	6702	6
MH-53J/M	92 FT 9 IN	7 FT 6 IN	9785	6458	6
C-32A (757)	155 FT 3 IN	11 FT 4 IN	16973	11202	7
E-8C (707)	152 FT 11 IN	11 FT 8 IN	16768	11067	7
E-3B (707)	146 FT 6 IN	11 FT 8 IN	16065	10603	7
C-22B	133 FT 2 IN	11 FT 4 IN	14558	9609	7
RC-135U/V/W (707)	140 FT 6 IN	11 FT 8 IN	15407	10169	7
NFPA Category 8 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
B-1B	146 FT 0 IN	10 FT 0 IN	15768	10407	7
B-2	69 FT 0 IN	10 FT 0 IN	4554	3006	4
B-52H	158 FT 4 IN	12 FT 4 IN	17469	11529	7
KC-135 (707)	136 FT 3 IN	11 FT 2 IN	14873	9816	7
KC-46 (767)	159 FT 2 IN	16 FT 4 IN	18228	12030	7
C-17A	174 FT 0 IN	18 FT 0 IN	20184	13321	8
The aircraft highlighted in yellow above were moved to a higher airport category because of their fuel loads and the B-2's wing design.					
NFPA Category 9 Aircraft by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
KC-10A (DC-10)	181 FT 7 IN	19 FT 9 IN	21381	14111	8
E-4B (747)	231 FT 4 IN	11 FT 6.5 IN	25340	16724	9
VC-25 (747)	231 FT 10 IN	20 FT 2 IN	27161	17926	9
The KC-10A aircraft highlighted in yellow was moved to a higher airport category because of its fuel load.					

NFPA Category 10 Aircraft by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
C-5A/B	247 FT 10 IN	23 FT 9 IN	30173	19914	10

US Army Q Factor Calculations by Type Aircraft and Category.

NFPA Category 1-4 Airports by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
T-41	26 FT 5 IN	4 FT 0 IN	1161	766	1
T-42A	28 FT 0 IN	3 FT 6 IN	1204	795	1
MQ-1C (RPA)	23 FT 0 IN	3 FT 7 IN	989	653	1
MQ-5B (RPA)	23 FT0 IN	4 FT 8 IN	1012	668	1
RQ-5 (RPA)	22 FT 9.6 IN	4 FT 8 IN	1012	668	1
U-10	30 FT 8 IN	3 FT 9 IN	1333	880	1
CESSNA 182	25 FT 2 IN	3 FT 8.5 IN	1075	710	1
MH-6	32 FT 7 IN	4 FT 7 IN	1452	959	2
TH-67	31 FT 2 IN	6 FT 3.5 IN	1426	942	2
UH-72A	33 FT 4.8 IN	5 FT 8.4 IN	1507	995	2
U-8F	31 FT 6 IN	4 FT 0 IN	1376	908	2
U-9	35 FT 9.75 IN	5 FT 0 IN	1584	1046	2
UV-20A	36 FT 1.2 IN	3 FT 9.5 IN	1548	1022	2
U/RU-21	35 FT 6 IN	4 FT 6 IN	1584	1046	2
CESSNA Caravan	37 FT 7.2 IN	3 FT 8.5 IN	1634	1078	2
AH-1S	57 FT 3 IN	8 FT 0 IN	3047	2011	3
AH-64A	49 FT 1 IN	6 FT 3 IN	2597	1714	3
AH-64D	49 FT 1 IN	6 FT 3 IN	2597	1714	3
CH-47	50 FT 0 IN	11 FT 11 IN	2700	1782	3
HH-60	50 FT .75IN	7 FT 9 IN	2700	1782	3
OH-58A/C	41 FT 2.4 IN	7 FT 4 IN	2268	1497	3
OH-58D	41 FT 2.4 IN	7 FT 4 IN	2268	1497	3
UH-1	57 FT 4 IN	8 FT 0 IN	3105	2049	3
UH-60A/L/M	50 FT .75IN	7 FT 9 IN	2700	1782	3
U-35A/C	48 FT 11 IN	5 FT 11 IN	2548	1682	3
C-12A/C	43 FT 10 IN	4 FT 6 IN	2244	1481	3
C-21	45 FT 1 IN	4 FT 11 IN	2295	1515	3
C-23A	58 FT 0 IN	6 FT 4 IN	3074	2029	3
OV-1	41 FT 0 IN	5 FT 0 IN	2091	1380	3
C-31A	77 FT 3.5 IN	5 FT 0 IN	4717	3114	4

NFPA Category 5 Airports by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
C-8A	79 FT 0 IN	8 FT 9 IN	7742	5110	5
C-20A/B/C/H	83 FT 2 IN	7 FT 4 IN	8183	5401	5
CH-54	88 FT 6 IN	19 FT 8 IN	8673	5724	5

NFPA Category 6 & 7 Airports by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
C-22B (727)	133 FT 2 IN	11 FT 8 IN	13034	8603	7
NFPA Category 8 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
N/A	N/A	N/A	N/A	N/A	N/A
NFPA Category 9 Aircraft by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
N/A	N/A	N/A	N/A	N/A	N/A
NFPA Category 10 Aircraft by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
N/A	N/A	N/A	N/A	N/A	N/A

US Navy/Marines Q Factor Calculations by Type Aircraft and Category

NFPA Category 1-4 Airports by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
T-34C	28 FT 8 IN	2 FT 10 IN	1189	785	1
T-2	38 FT 7 IN	5 FT 0 IN	1716	1133	2
T-44	35 FT 4 IN	4 FT 6 IN	1562	1031	2
AV-8B	46 FT 3.6 IN	5 FT 0 IN	2372	1565	3
F-35B	51 FT 5.16 IN	15 FT 0 IN	3142	2074	3

F-35C	51 FT 5.16 IN	15 FT 6 IN	3193	2108	3
F-5E/F	51 FT 7 IN	5 FT 0 IN	2652	1751	3
F-18C/D	56 FT 0 IN	5 FT 0 IN	2856	1885	3
T-39	46 FT 8 IN	5 FT 3 IN	2444	1613	3
T-45A	39 FT 3 IN	5 FT 0 IN	2002	1322	3
TA-4	42 FT 7 IN	4 FT 0 IN	2150	1419	3
AH-1W	44 FT 7 IN	7 FT 3 IN	2430	1604	3
H-2	52 FT 7 IN	7 FT 10 IN	2862	1889	3
H-46	45 FT 8 IN	7 FT 3 IN	2484	1640	3
UH/SH-3	54 FT 9 IN	6 FT 6 IN	2915	1924	3
TH-57	39 FT 8 IN	4 FT 2 IN	2040	1347	3
UH-1H	57 FT 1 IN	8 FT 7 IN	3135	2070	3
MV-22	57 FT 4 IN	5 FT 11 IN	2980	1967	3
S-3	53 FT 4 IN	7 FT 2 IN	2880	1901	3
C-2	56 FT 10 IN	7 FT 4 IN	3078	2032	3
UC-12	43 FT 9 IN	4 FT 6 IN	2244	1482	3
E-2	56 FT 4 IN	7 FT 4 IN	3042	2008	3
UC-26	59 FT 4.25 IN	5 FT 2 IN	3690	2436	4
F-18 E/F	60 FT 3.5 IN	5 FT 0 IN	3721	2456	4
HH-60H/SH-60	64 FT 8 IN	7 FT 9 IN	4138	2732	4
NFPA Category 5 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
C-20 (Gulfstream III)	83 FT 1 IN	7 FT 4 IN	8134	5369	5
H-53D	88 FT 6 IN	8 FT 10 IN	8673	5725	5
NFPA Category 6 & 7 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
P-3	116 FT 10 IN	10 FT 10 IN	11466	7568	6
C-9 (DC-9)	104 FT 4.8 IN	9 FT 5 IN	10290	6792	6
C-130	97 FT 9 IN	10 FT 3 IN	9604	6339	6
H-53E	99 FT 0.5 IN	8 FT 10 IN	9702	6404	6
E-6B (707)	152 FT 11 IN	11 FT 8 IN	14994	9896	7
EA-6B	159 FT 10 IN	11 FT 8 IN	15680	10349	7
NFPA Category 8 Airports by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
N/A	N/A	N/A	N/A	N/A	N/A

NFPA Category 9 Aircraft by Aircraft Type	Fuselage Length	Fuselage Width	TCA Sq Ft L x (K+W)	PCA Sq Ft .66 x TCA	NFPA Airport Cat Table 4.3.1
N/A	N/A	N/A	N/A	N/A	N/A
NFPA Category 10 Aircraft by Aircraft Type	**Fuselage Length**	**Fuselage Width**	**TCA Sq Ft L x (K+W)**	**PCA Sq Ft .66 x TCA**	**NFPA Airport Cat Table 4.3.1**
N/A	N/A	N/A	N/A	N/A	N/A